

Exploratory Data Analysis (EDA) Insights Report

Correlation Heatmap :

The heatmap shows several important relationships:

Strong Positive Correlations:

- ✓ **Close, Open, High, Low, Adjusted Close**
→ Highly correlated (0.9–1.0), expected for OHLC data.
- ✓ **SMA20, SMA50, SMA200 with price**
→ Shows long-term trend dependency.
→ SMA200 correlation slightly lower due to smoothing.
- ✓ **EMA12 ↔ EMA26 ↔ MACD**
→ Logical, since $MACD = EMA12 - EMA26$.
- ✓ **BB Upper / Lower / Middle with price**
→ Bollinger bands follow price tightly.

Moderate Correlations:

- ✓ **RSI(14) ↔ Momentum(10/20)**
→ RSI is a momentum oscillator → expected overlap.
- ✓ **ATR(14) ↔ True Range & Volatility20**
→ Measures of price range and volatility.
- ✓ **MACD Histogram ↔ MACD**
→ Histogram is the difference between MACD and signal.

Low Correlations:

Volume metrics vs Price

→ Low correlation indicates volume is not strongly tied to price level (good for model diversification).

Future Return 5d vs Indicators

- Weak linear correlation is normal; financial markets are nonlinear.
 - Suggests ML will perform better with tree-based models or sequence models.
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Trend Analysis:

Price Trends

- **STK001:**
Strong upward long-term trend from mid-2022 through 2024.
Suggests momentum behaviour, good for trend-following models.
- **STK002:**
Clear long-term decline until mid-2023, followed by mild recovery.

Behaves like the mean-reversion candidate.

- **STK003:**
Large spike mid-2023, forming a parabolic move with correction.
Volatile, event-driven behaviour.

RSI Insights

Across all stocks:

- Regular oscillations between **30 (oversold)** and **70 (overbought)**.
- RSI rarely stays in extreme zones → suggests balanced mean-reversion.

MACD Insights

- MACD/Signal crossovers show major trend shifts.
- STK001 shows extended bullish phases (large swings above zero).
- STK002 shows more time near the zero line → weak trends.
- STK003 shows strong but short-lived momentum surges.

Bollinger Band Behaviour

- STK001 price rides upper band during uptrend, this shows a strong bullish momentum.
- STK002 touches lower bands frequently, this reflects a persistent downtrend.
- STK003 expands band width significantly during volatility events.

Conclusion:

Each stock has different trend regimes, which is valuable for a diversified modeling dataset.